Computer Science & IT



C Programming

Data Types and Operators

Lecture No. 02



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Recap of Previous Lecture

integer



- topic — Problem Solving
- topic — Character
- topic — Range of Character
- topic
- topic

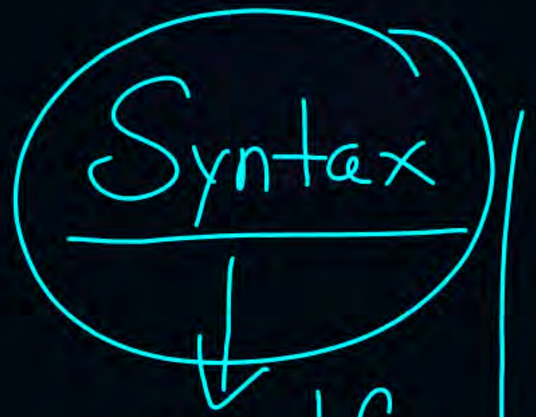
Slide

topics to be Covered



- topic Integer Signed
- topic Integer Unsigned
- topic Character
- topic Character Signed
- topic Character UnSigned

Slide



Learn

programming

printf("%d", a)

Slide



topic : Question

```c
#include <stdio.h>
int main(){
        short int a = 10;
        printf("%hd", a);
        printf("%hu", a);
        return 0;
}
```

short int a = 10 ← 2B

unsigned short

2B

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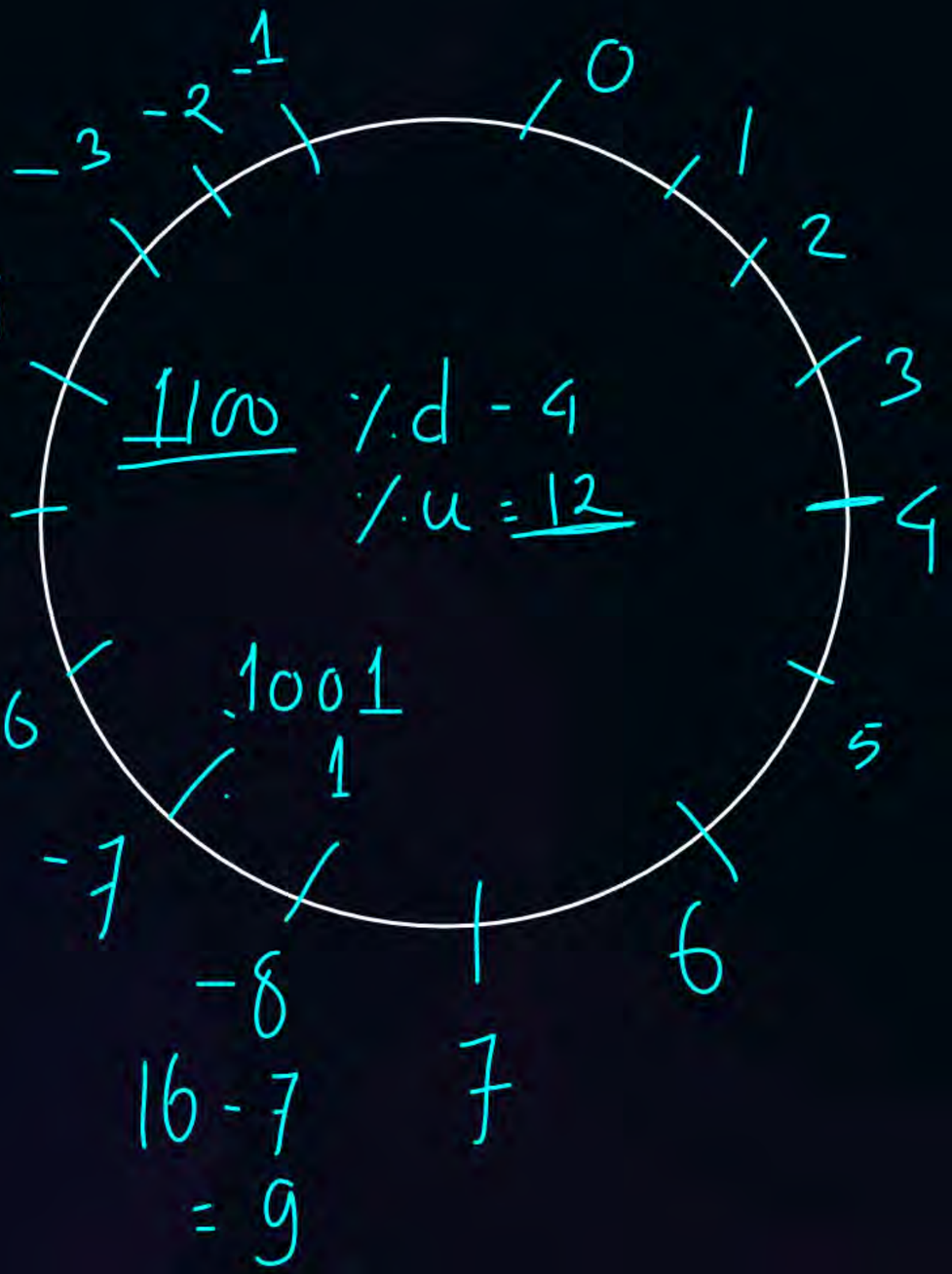
ot

igned

$2^{n-1} - 1$

$2^{4-1} - 1$

7



1100 %d - 4

%u = 12

1001

1

16 - 7

= 9

4 bit ✓

8 bit ← char

16 bits

a = 4

```
printf("%d", a)
printf("%u", a)
```

a = -4 ←

%d ← ✓

0100

↓

1011

1

1100 ✓

%u

formula

4 bits

$2^4 = 16$

-4 ⟶ %d — 4

%u 16 - 4 = 12

Unsigned value - No. o

Slide

Unsigned value of Negative No.

No. of bits = short int = 2B = 2 × 8 = 16

$2^{16} - 10 = 65536 - 10 = 65526$

hu ← 2B a unsigned.

$2^{\text{No. of bits}} - 10$

Negative

Slide



topic : Question

```
#include <stdio.h>
int main(){
        short int a = -10;
        printf("%hd", a);
        printf("%hu", a);
        return 0;
}
```

-10

-10

Size = 16 bits

$2^{16} - 10 = 65536 - 10$

65526

unsigned

Slide



topic : Question

```
#include <stdio.h>
int main(){
        short int a = -20;
        printf("%hd", a);
        printf("%hu", a);
        return 0;
}
```

A. 20 & 65516

B. -20 & 65516

C. -20 & 20

D. 20 & -65516

$2^{16} - 20$

$65536 - 20$

65516



topic : Question

```
#include <stdio.h>
int main(){
        short int a = -48;
        printf("%hd", a);
        printf("%hu", a);
        return 0;
}
```

2B

-48

$2^{16} - 48$

65536

$$\begin{array}{r} 48 \\ - \quad 1\,1 \\ \hline 65488 \end{array}$$

A. -48 & 65488

B. -48 & -65488

C. 48 & 65488

D. -48 & -65516

Slide



topic : Question

```
#include <stdio.h>
int main(){
        short int a = -48;
        printf("%hd", a);
        printf("%hu", a);
        return 0;
}
```

A. -48 & 65488

B. -48 & -65488

C. 48 & 65488

D. -48 & -65516

Slide



Homework

```
#include <stdio.h>
int main(){
        short int a = 32770;
        printf("%hd", a);
        printf("%hu", a);
        return 0;
}
```

A. -32766 & 32770

B. 32766 & 32770

C. 32770 & 32770

D. -32766 & 32776

Size - 4 Byte : 32 bits

$2^{32} - 48$

Slide



Character

Size - 1 Byte ✓

unsigned / Singed ✓

ASCII Code

char ch = 'a'; Singe quote

↑ Datatype — Name — Initialize

decimal



ASCII value

0…9	48…57
A…Z	65…90
a…z	97…122



Sign Character

```c
#include <stdio.h>
#include <stdio.h>
int main(){
        char ch1 = 'a';
        char ch2 = 'z';
        printf("%c\n", ch1);
        printf("%d\n", ch1);
        printf("%c\n", ch2);
        printf("%d", ch2);
return 0;
}
```

a, b, c, d, e. . . . z.

%c ⟷ %c

Singned :

a - 97 ✓

b = 98

c - 99

d - 10

z - 122 ✓

A B ASCII table

65 66

01000001 ✓

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Sign Character

A - 65

Z - 90

```
#include <stdio.h>
#include <stdio.h>
int main(){
        char ch1 = 'A';
        char ch2 = 'Z';
        printf("%c\n", ch1);
        printf("%d\n", ch1);
        printf("%c\n", ch2);
        printf("%d", ch2);
return 0;
}
```

Slide



Sign Character

```c
#include <stdio.h>
#include <stdio.h>
int main(){
        char ch1 = '0';
        char ch2 = '9';
        printf("%c\n", ch1);
        printf("%d\n", ch1);
        printf("%c\n", ch2);
        printf("%d", ch2);
return 0;
}
```

ASCII
48 - '0'
57 - '9'

Symbols

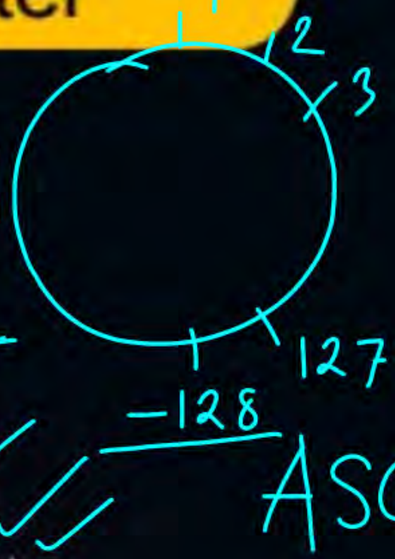
ASCII table (7 bits)

Extended ASCII table (8 bits)

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Sign Character



what it will print

```
#include <stdio.h>
int main(){
    char a = 128;
    char c = -128;
    printf("%c\n", a);
    printf("%d\n", a);
    printf("%c\n", c);
    printf("%d", c);
    return 0;
}
```

HW

8 bits

ASCII code

7 bits

Extended ASCII

8 bits

problem 1 Hrs

1 Hrs ← problem.

@ character assigning Numeric value

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8 bits

-2^{8-1} to $2^{8-1}-1$

-128 to 127

"%d"



Singed No.

Slide

8 bit

0 — $2^8 - 1$

0 - 255



unsigned

Char

printf("%u"

Slide



Signed Char

```
#include <stdio.h>
int main() {
char ch = -134;
printf("%c", ch);
return 0;
}
```



UnSigned value

(d)
Unsigned value for –134= 256 – 134 = 122.
Hence, ‘z’ is printed.



Signed Char

```
#include <stdio.h>
int main() {
char ch = -139;
printf("%c", ch);
return 0;
}
```

The output is-

(a) u

(b) Garbage

(c) Overflow warning

(d) U



topic : variable

Consider the following program:

```
#include <stdio.h>
int main()
{
char ch=119;
ch=ch+6;
printf("%d", ch);
return 0;
}
```

The output is- _____.

Slide



&pic : Reserve keyswords

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while



2 mins Summary

&pic

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THANK - YOU

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